Anatomy of young dicotyledonous stem

It differentiates into dermal, fundamental and vascular system

1- Demal system: It consists of one layer of barrel- sha epidermal cells covered with cuticle.

- It has stomata and hairs.

2- fundamental system:

A- Cortex: It may consist of:

- parenchyma (may contain plastids)
- Collenchyma (as cylinder or strips)
- Fibres.

B- Endodermis: It seperates the cortex from the vascular tissue. It is not well developed.

- The innermost layer of cortex contains starch and kno as starch layer.

3- Vascular system.

a- Pericycle (protophloem fibres): forms cap on vascular bundle.

b- Vascular bundles (V.B.):

- It is open due to presence of cambium.
- V.Bs. vary in number
  - * In square stem, one is present at each corner.
  - ** In rounded stems, they are 15 or 20 bundles in one o two rings.
- V.Bs. may be

* collateral: phloem is external to xylem.

** bicollateral: phloem on both sides of xylem.

i- Phloem: It differentiates to:

a- Protophloem: composed of sieve tubes and companion cells. The sieve elements are narrow.

b- metaphloem: composes of sieve tubes, companion cells and parenchyma. The sieve elements are wider.

ii- Cambium:

- It is meristematic with thin walled cells.
- It lies between xylem and phloem and known as vascular cambium. Interfasicular cambium joins the fasicular ones during 2ry growth.

iii- Xylem:

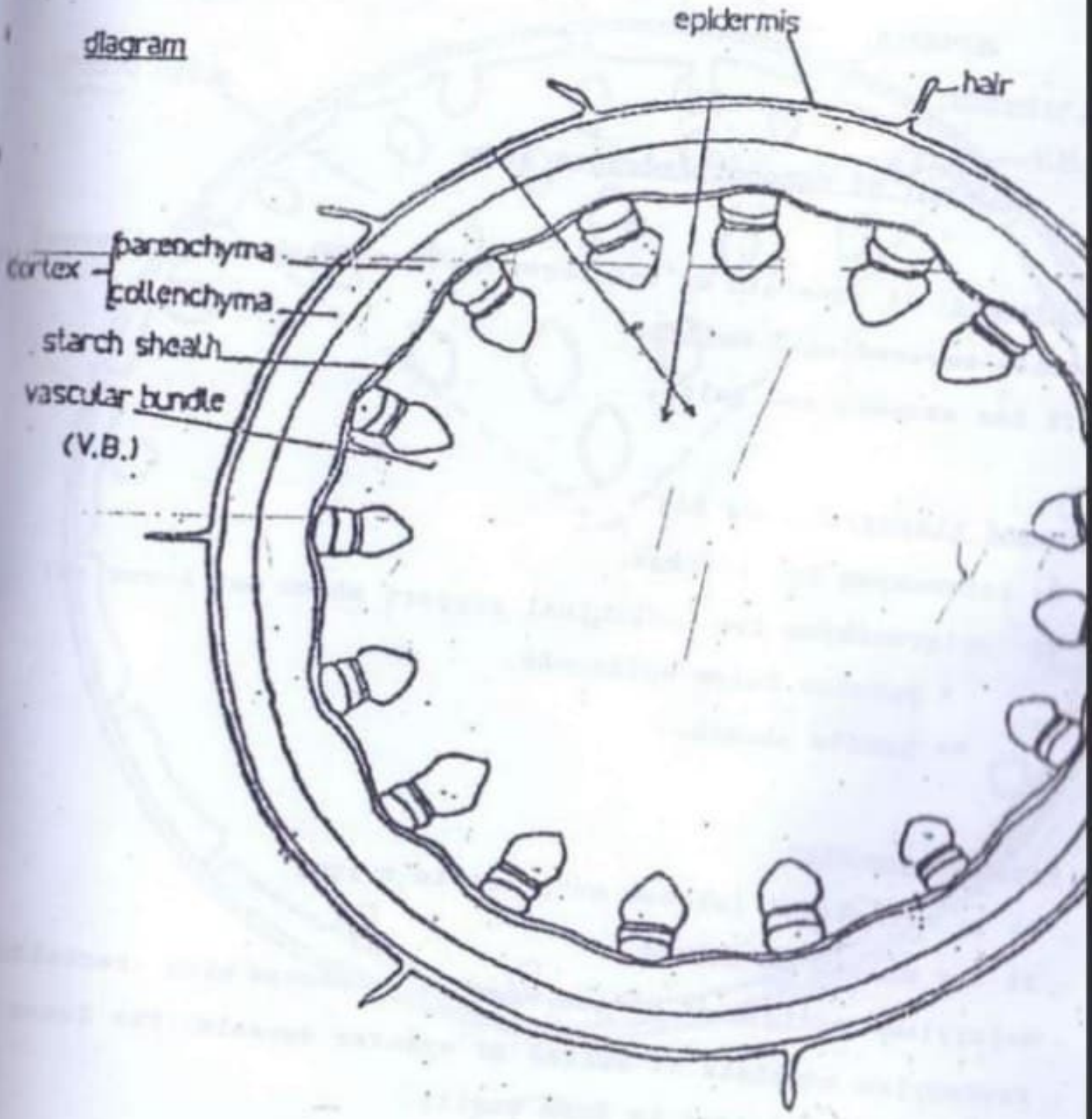
- Composes of protoxylem, metaxylem and paranchyma.
- Protoxylem directs inwards (endarch)

Pith:

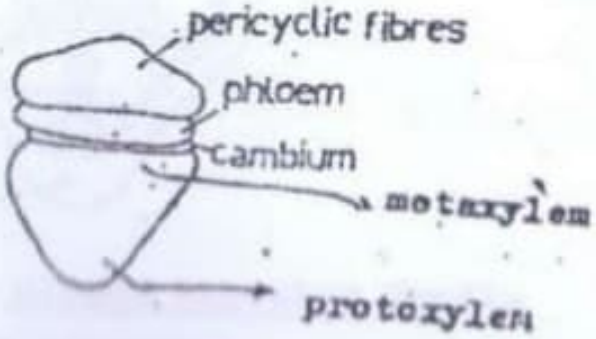
- It is parenchyma cells with intercellular spaces.
- They are larger in the centre.
- Somtimes the pith cells are ruptured and the stem is hollow.
- It occurs also between V.Bs.

T.S. in young dicot stem

diagram



opened V.B.



Anatomy of monocotyledonous stem

1- Epidermis: It consists of one layer of barrel- shaped epidermal ~~cells covered with cuticle~~

- It has stomata and hairs.

2- Ground tissue: It may be:

i- parenchyma for storage.

ii- sclerenchyma for mechanical support which may occur as:

* patches below epidermis.

** bundle sheath.

3- Vascular bundle:

- It is collateral (phloem external to xylem)
- It has sheath of fibres.
- Metaxylem consists of pitted vessels [illegible] nected with tracheids.
- Protoxylem consists of spiral or annular vessels. The inner protoxylem dissolves to form cavity.
- There is no cambium (V.B. is closed).

General characters of stems:

1- The presence of epidermis covered with cuticle.

2- The xylem and phloem are arranged on the same radius.

3- The protoxylem vessels are directed inwards i.e. endarch.

Characters of dicot stems:

1- The ground tissue is differentiated into a wide pith and narrow cortex.

2- The presence of starch sheath and pericyclic fibres.

3- The vascular bundles are arranged in one cycle.

4- The V.Bs. are open due to presence of cambium.

5- The xylem vessels are arranged in rows.

6- The phloem consists of:

a- sieve tubes.

b- companion cells.

c- parenchyma.

Characters of monocot stems:

1- The ground tissue is undifferentiated into cortex and pith.

2- The absence of starch sheath and pericyclic fibres.

3- The V.Bs. are scattered in the ground tissue.

4- The V.Bs. are closed due to absence of cambium

5- The xylem vessels are V- or Y- shaped.

6- The phloem is regular and consists of sieve tubes and companion cells only.

Anatomy of The roots

1- Epidermal and piliferous layer:

- It is thin walled cells without cuticle.
- In region of hair, the epidermis is known as piliferous layer.
- In moist, well aerated soil, the root hair is densed than less aerated one.
- Hairs are used for absorption.
- The epidermis is followed by suberized cells used for protection and known as exodermis.
- The exodermis has thin walled passage cells.

2- Cortex:

- It is wider in root than stem
- It is composed of parenchyma and sclerencyma. No collenchyma in roots.

3- Endodermis:

- It is four- sided cells.
- The radial walls have suberized bands known as casparian strip which may be: a- in monocots: on tangential (side) and radial (inner) walls (U- shaped) or in all walls (O- shaped).

  b- in dicots on tangential walls
- In monocot, it contains passage cells.

4- Pericycle:

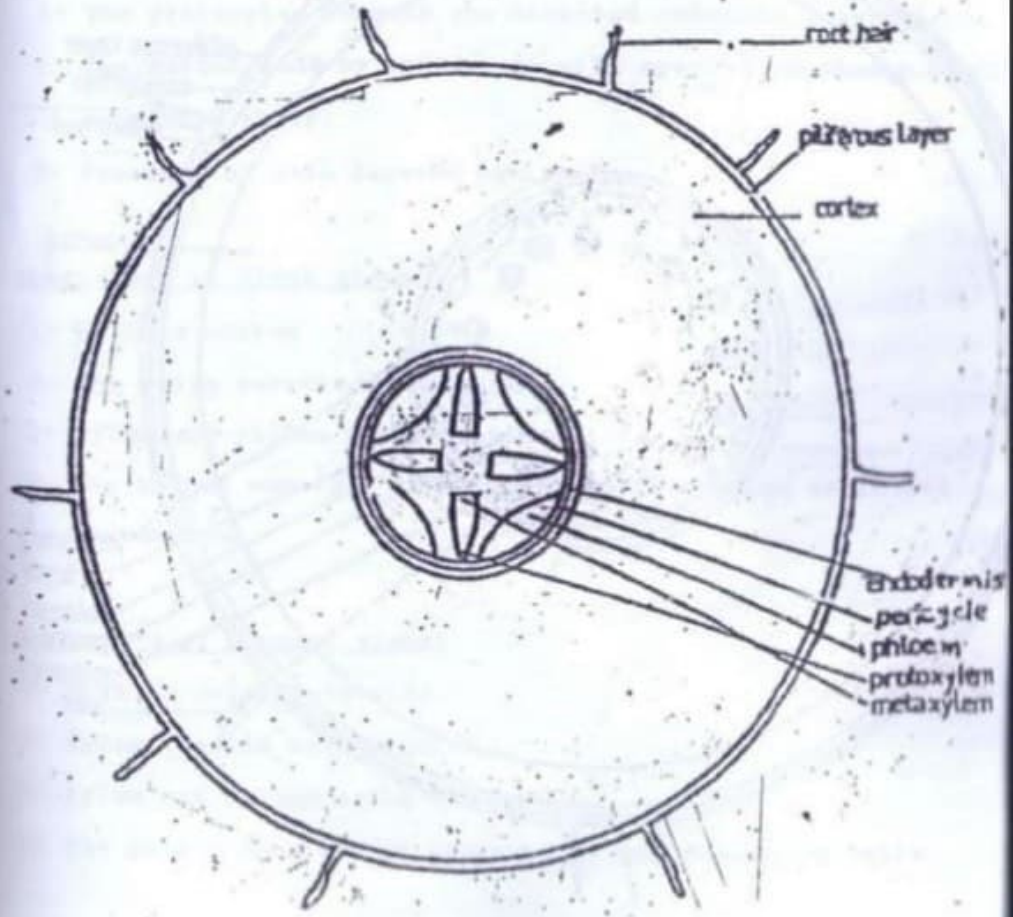
- It is parenchyma cells following the endodermis.
- It is used for:
  - a- production of lateral branches.
  - b- production of cork cambium in older roots.

5- Vascular cylinder:

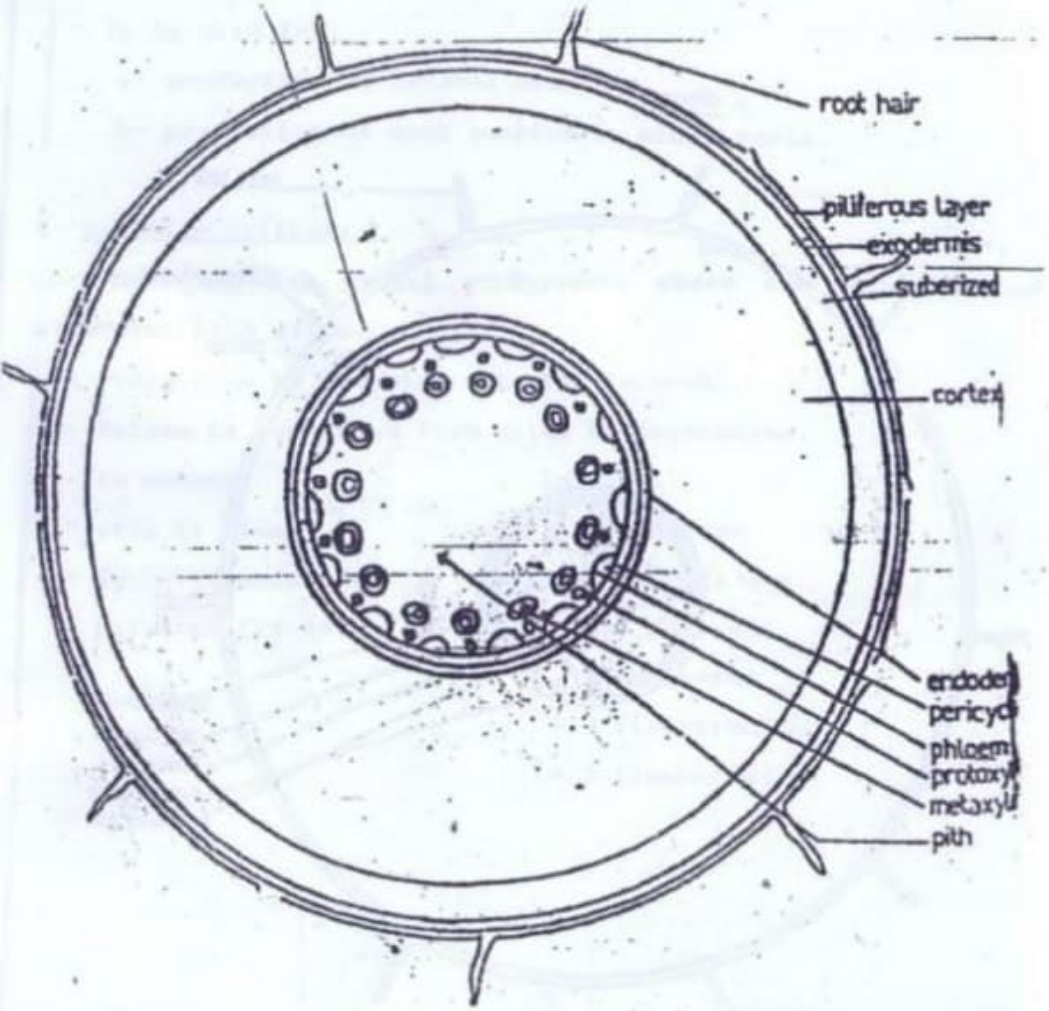
- The root has radial arrangement where the phloem strands alternate with xylem.
- Protoxylem is directed outwards (exarch).
- Phloem is seperated from xylem by parenchyma.

In monocot	in dicot
* pith is wider	* pith is narrow
* Xylem strands are polyarch (15-20)	* Xylem strands are:
	- two (diarch)
	- 3 (triarch)
	- 4 (tetrarch)
	- 5 (pentarch).

diagram



T.S. in monocot Stem.



General characters of roots:

1- Presence of piliferous layer provided with root hairs an without cuticle.

2- The xylem and phloem are arranged on different radii.

3- The protoxylem vessels are directed outwards (exarch).

4- The cortex ends by endodermis with special thickness of casparian strip.

5- Presence of chin layered pericycle.

Characters of dicot stems.

1- Pith is narrow

2- The xylem vessels are angular.

3- Xylem and phloem archs not exceed 8.

4- The phloem consists of sieve tubes, companion cells and parenchyma.

Characters of monocot stems:

1- Pith is relatively wide.

2- Xylem vessels are circular.

3- Xylem and phloem archs are > 8.

4- The phloem consists of sieve tubes and companion cells.

Anatomy of leaf

A cross section of leaf blade shows the following main systems:

1- Epidermis:

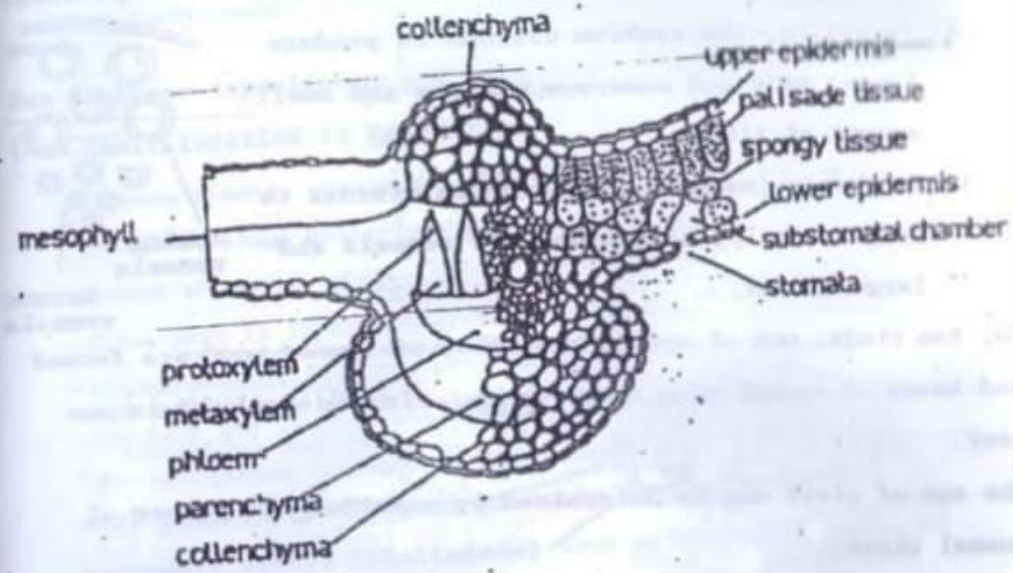
- It is barrel- shaped consisting of:
  - a- one ore more layer of cells
  - b- has cuticle of varying thickness
  - c- has no chloroplast except water plants and guard cells.
  - d- It has: i- guard cells and stomata.
    ii- hairs.
  - e- Some cells may become swollen and responsible for leaf folding.

2- Mesophyll:

- It does not differentiate in monocot while in dicot, it differentiates to:
  - a- palisade Tissue: It is 1 on epidermis
    - It is arranged in one or more layers, which may be in one side (in dorsiventral leaf) or both sides (in isobilateral leaf).
    - It contains more chloroplasts, so it is deeper green than spongy in dorsiventral leaves.
  - b- Spongy tissue: It has wider intercellular spaces and used for gaseous exchange.

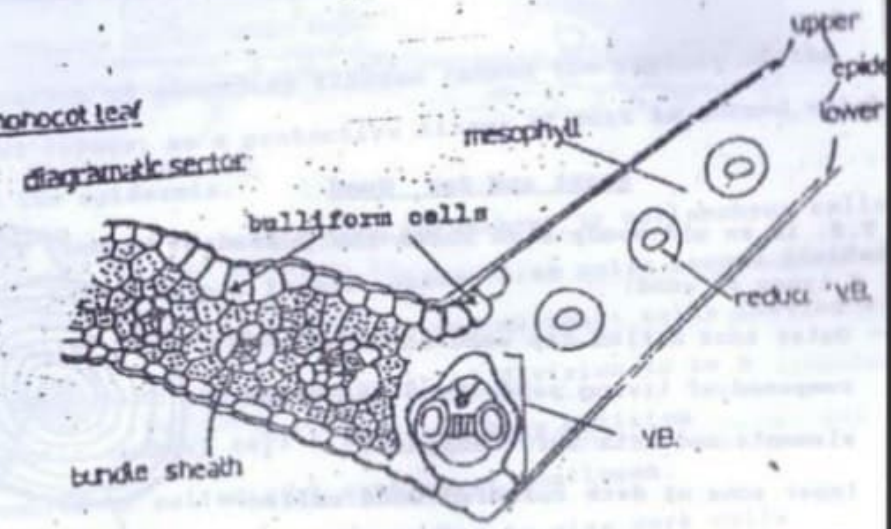
T.S. in dicot leaf

diagramatic sector



T.S. in monocot leaf

diagramatic sector

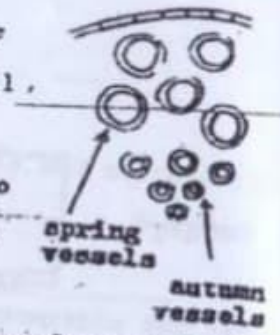


Annual Rings

In cold and temperate regions, the activity of the cambium varies from one season to the other.

a- In spring, the cambium divides to produce large, wide and numerous vessels and small amount of fibres.

b- In summer (autumn), the cambium divides to produce small, narrow and few vessels and large fibres.



So, two rings, one of spring and other of summer wood are formed and known as annual ring. Each annual ring is produced in one year.

The age of plant can be determined by counting the number of annual rings.

Heart and Sap Wood

A T.S. in an old woody stem shows the presence of 2 types of wood:

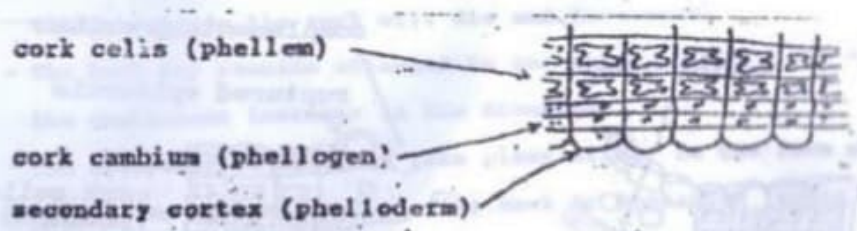
a- Outer zone called sap wood which is composed of living parenchyma and other elements and acts for transport.

b- Inner zone of dark coloured wood called the heart wood. Heart wood consists of dead elements which are dark due to the

deposition of resins, gums, pigments tannins and other substances which are antiseptic against decay.

- These depositions increase the weight and durability of the heart wood.
- Due to these depositions and formation of tyloses, heart wood lose their function in conduction.

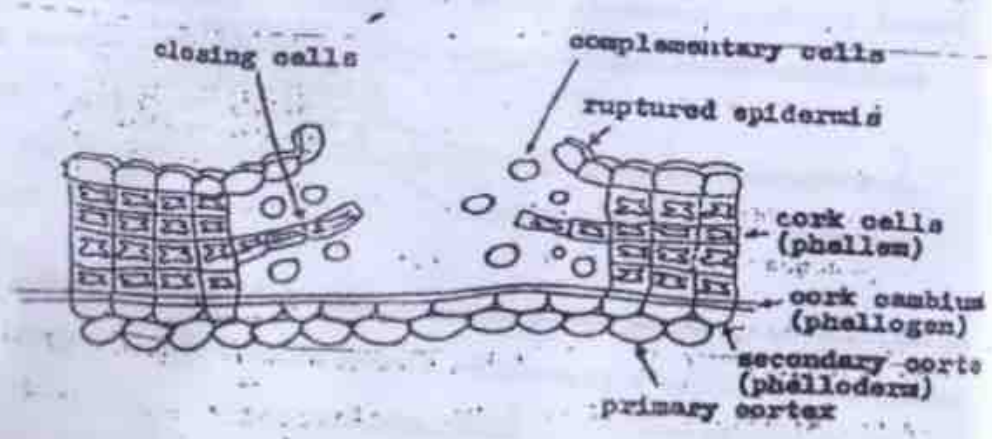
Periderm



- The formation of secondary tissues causes the rupture of the outermost layers, so a protective tissue of cork is formed which replace the epidermis.
- This cork cambium is formed from parenchyma or collenchyma cells of the cortex or deeper layers, where these cells become divided by 2 parallel walls into an outer and an inner cells leaving a middle cell which regain the power of division to be a meristematic cambial cell i.e. it is 2 ry meristem
- This cambium is called cork cambium or phellogen.
- The phellogen divides to the exterior to give cork cells (phellem) and to inside to give parenchyma or collenchyma tissue called the phelloderm or 2 ry cortex.

- In T.S. the cork cells are: - rectangular - have no intercellular spaces - with suberized cell walls - non-living when mature - impermeable

Lenticels



- The formed cork cut the diffusion of gases in and out the tissues below it, so special ventilating regions are formed known as lenticels.
- In the region of the lenticel, phellogen produces large thin walled cells with intercellular spaces which allow the diffusion of gases and these cells are called complementary cells.
- In some plants, in winter, phellogen produces a closing tissue

instead of giving rise to complementary cells.

- In spring, new complementary cells are formed causing the rupture of the closing tissue.
- Lenticels appear as slit in surface view.

Bark

- In some plants, phellogen remains active for years while in others it stops activity and a new phellogen is formed in deeper layers of cortex giving deeper new cork.
- Since cork is impermeable to water and nutrients, so any living tissue outside the cork will die and form bark.
- The bark may remains attached to the plant or may split due to the continuous increase in the diameter.
- The splitting of bark may take place either in the form of continuous ring and called ring bark or separated layers or scales and called scale bark.
- The bark may contain alkaloids, volatile oils or tannins